SOW for LISA Laser Reliability Subject Matter Expert (SME) Support

Dr. Scott Merritt (Sole Proprietor, dba Merritt Analytics)

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Background
The gravitational wave (GW) space mission Laser Interferometer Space Antenna (LISA) has been chosen for the European Space Agency (ESA) L3 GW astronomy opportunity and NASA intends to contribute instruments to the mission. LISA's eventual flight will open a spectacular new window on the universe, using gravitational waves to reveal the physics and astronomy associated with the merger of massive black hole systems. The backbone of LISA is a highly stable laser of ~2 W power, which enables the picometer interferometry necessary to record the passage of a gravitational wave. Such a laser is listed as a top priority of the LISA project under Code 460.

One of the more stringent requirements for the LISA laser system is lifetime. The lifetime requirement breakdown for the laser system is as follows

- 6 years comprising 1 year for on-ground testing plus 5 years (TBC) of storage
- 1.5 years OFF state in an operational environment (cruise phase)
- 5 (TBC) years continuous operation in nominal science mode (nominal mission lifetime)
- 6 (Goal) additional years continuous operation in nominal science mode (extended mission lifetime)

An effective reliability program must include

- a sound plan for life-testing components and subsystems
- understanding the physics of failure of relatively high-risk components so the initial probability of failure is minimized.
- reduce the final probability of failure, e.g. conditioned on passing screening tests ("screens") at the component level.
- guide the design of the screening protocols then provide proof (or disproof) of the efficacy of the screens, procedures, and choice of operating conditions.

Scope
This task is for a subject matter expert (SME) working closely with GSFC TM to establish a reliability program for the NASA GSFC in-house laser development for LISA. The laser system for LISA under development includes – laser head (LH), frequency reference system (FRS) and the power monitoring detectors (PMON). The objectives of this task are to develop reliability models and implement design of experiment (DOE)

trials to address the lifetime concerns of spaceborne lasers from components to subsystems and systems that are applicable to the LISA mission; provide guidance and implementation of DOEs to address lifetime concerns; analyze lifetest data and update reliability models; develop future lifetest parameters on components, subsystems and systems, and provide reports and presentations for the LISA program.

Requirements

- Travel domestically and internationally to visit vendors' sites to evaluate individual vendor's reliability program.
- Attend and present at domestic or international conferences to meet with other reliability engineers and SMEs for technical interchange meetings that will support and benefit the LISA laser development.

Deliverables or Delivery Schedule

#	Milestones	Dates (# weeks after receipt of order (ARO))
1	Trip reports	Throughout POP, within 2 weeks of completion of travel
2	Delivery and continue update of reliability models for laser components, laser subsystems, and laser system	Quarterly
3	Update LISA laser reliability plan	No later than (NLT) 50 weeks ARO
4	Publish journal paper	NLT 50 weeks ARO

Government-Furnished Equipment and Government-Furnished Information

- Test equipment necessary for development of reliability test stations.

Place of Performance

- Majority of the work will be performed at contractor's site.
- Travel to support LISA laser reliability program

Period of Performance

- **4/15/2025 through 4/14/2026.**